

STANFORD UNIVERSITY
Department of Electrical Engineering

Lab 2: Schmitt trigger

Due: Friday, 4 May 2026

Reading: Notes on function generators; schematics of HP and Tektronix function generators

Assignment: Design, build, and demonstrate a Schmitt trigger (“hysteresis switch”) to be used as part of your function generator. The trigger should meet the following specifications for an input triangle wave frequency up to 2MHz:

Input resistance: $>1\text{k}\Omega$ (the tri-wave buffer will have to drive this)

Trigger thresholds: $+0.5\text{V}/-0.5\text{V}$ nominal; each independently adjustable by $> \pm 0.15\text{V}$

Square wave output(s): 1V peak-to-peak, centered about zero volts.

Square wave output rise/fall time: $< 20\text{ns}$ for output that drives your final amplifier from Lab 1, when loaded by same; $< 10\text{ns}$ for output that drives the switch core, when loaded with oscilloscope probe ($\sim 15\text{pF}$). (If you provide only one output overall, then spec is $< 10\text{ns}$ for that output when fully loaded as in final system.)

Square wave output overshoot: $< 2\%$

Trigger latency: $< 10\text{ns}$ (See note at end)

Supply voltage: $\pm 12\text{V}$ (as always for this course)

Allowed components: Same complement as for the output amplifier. You may also consider using the 1N5711/1N6263 small-signal Schottky diodes we have stocked in the lab if you do not wish to use the 1N4148/1N914 junction diode. Just keep in mind that Schottkys have considerably higher junction capacitance per unit current than do junction devices, and generally higher reverse leakage as well. Make sure that you have studied the relevant datasheets to assess suitability. Don’t assume that Schottkys are automatically better.

Some examples to get you thinking: The hysteretic switch of the Tektronix CFG250 function generator (see handout) is a good transistor-level circuit to inspire your own design. The HP circuit (again, see the handouts) may also stimulate some ideas. In any case, don’t merely attempt to copy these schematics, as your components and specs differ, and at least one circuit diagram is a simplified version of the actual implementation. You are expected to demonstrate design creativity and perhaps even outdo them both!

Measuring trigger latency: Recall from the system block diagram that the Schmitt trigger forms a loop with an integrator to generate square and triangle waves, respectively. The triangle wave’s amplitude is determined by the Schmitt trigger thresholds.

As discussed in class there is inevitably a delay from when the ramp crosses the Schmitt trigger threshold to when the output changes state, so the triangle wave keeps ramping for a while. At low frequencies, the slope of the ramp is small and so a given delay induces negligible voltage overshoot. However, at high frequencies, the slope of the ramp is high and that same delay may cause the ramp to overshoot significantly beyond the desired trigger points before the integrator turns around. The nonlinear network that converts a triangle into a sine works correctly only at a specific amplitude of the input triangle wave, so large variations in the amplitude of the input triangle wave will increase sine distortion to unacceptable levels.

The relevant delay is that between the time the input ramp crosses the Schmitt trigger threshold and when the square wave output that is driving the 4-diode switch core changes state. Here's one way to determine this latency parameter:

- 1) First, precisely measure the positive and negative thresholds of your Schmitt trigger by applying a slow triangle wave to the input and noting the input voltage at which the square wave toggles.
 - 2) Apply a high frequency triangle wave to the input of your Schmitt trigger and measure the delay between when the ramp crosses the trigger levels established in 1) and when the square wave toggles. Other measurement methods are certainly possible, and you are welcome to use whatever method you prefer. The formal specification is $< 10\text{ns}$, but you should aim for a delay that is a small fraction of the shortest period signal you wish to generate. As a crude rule of thumb, it is desirable to have a latency no larger than 10% of the half-period of the highest frequency to be supported. Keep in mind that the rise and fall times of your square wave will contribute measurably to this latency.
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